

ThinkFree Finance

ThinkFree is a modular financial intelligence system built with Python. It combines stock data scraping, technical analysis, AI-based summarization, article clustering, GPT economic reasoning, and profile-based investment strategies.

Modules Overview

1. Data Collection & Preparation

- `phase2_data_scraping.py` : Fetches article data from Finhub API
- `prepare_qlib_data.py` : Prepares and formats technical indicators for Qlib
- `phase_2_q_lib.py` : Integrates data into Qlib-compatible format

2. Signal Generation

- `phase_3_signal.py` : Calculates RSI, MACD, and crossover signals for all tickers

3. Backtesting (Phase 4)

- `phase_4_backtrader.py` : Backtests strategy logic using Backtrader
- Strategy: RSI < 30 + MACD crossover to buy; RSI > 70 or MACD crossover down to sell
- Outputs Sharpe ratio, win rate, average duration, and drawdown

4. Article Intelligence

- `article_clustering.py` : Clusters articles using sentence embeddings + KMeans
- `GPT_article_clustering_summary.py` : Summarizes each cluster for GPT inference
- Uses VADER for sentiment + regular expressions for extracting economic cues

5. User Personalization

- `# phase1_user_personalization.py` : Builds 12 user profiles based on age and risk:
- Profiles include: `A1_<25_R1_Cons`, `A4_50+_R3_High`, etc.
- `user_profile.json` : Current user profile used in matching stocks or strategies

6. Economic Reasoning Engine (Phase 6)

- Parses economic textbooks and Investopedia articles using FAISS
- Matches current indicators with embedded economic knowledge

7. Historical Pattern Correlation (Phase 7)

- Analyzes how current macro indicators compare to historical events
- Surfaces high-performing sectors from similar past conditions

How To Set Up On Another Computer

1. Clone the Repo

```
git clone https://github.com/allanaziz581/ThinkFree.git
cd ThinkFree
```

2. Install Dependencies

Create and activate a virtual environment:

```
python -m venv venv
source venv/bin/activate # or venv\Scripts\activate on Windows
```

Install packages:

```
pip install -r requirements.txt
```

If you don't have `requirements.txt`, install manually:

```
pip install pandas numpy yfinance backtrader sentence-transformers scikit-learn
matplotlib spacy vaderSentiment faiss-cpu
python -m spacy download en_core_web_sm
```

3. Setup Local Paths

Ensure these folders exist:

```
ThinkFree/
├─ qlib_data/formatted/
├─ Economic_Books/
└─ user_profile.json
```

If missing:

- Download `ta_analysis_detailed.csv` and `predictions.csv` into `qlib_data/formatted/`
- Provide or regenerate `user_profile.json`
- Optional: Rebuild `FAISS` index for economic books if needed

4. Run Example Modules

Run backtest:

```
python Module_2_Technical_Analysis/phase_4_backtrader.py --tickers AAPL TSLA
```

Run clustering:

```
python article_clustering.py
```

Project Status

- Fully modular with GPT-ready output
- Profile personalization fully integrated
- Historical correlation + economic logic modules structured but expandable

Future Improvements

- Real-time dashboard UI (Streamlit or React)
- GPT-4 API integration for personalized financial summaries
- Plug-and-play sector model training
- Recession Indicator Engine (Phase 13)

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